

Sumit Raj

AI/ML Engineer

CONTACT

Location: Patna, India **Phone:** +91 9472441137 **Email:** info.sr0909@gmail.com

LinkedIn: linkedin.com/in/er-sumit-raj- **GitHub:** github.com/sumit966

Portfolio: sumit966-github-io.vercel.app

SKILLS

Languages: Python, SQL

Machine Learning and Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, CNN, RNN, LSTM

Computer Vision: OpenCV, YOLOv8, Haar Cascade, Object Detection, Image Classification

GenAI and NLP: LLMs, RAG, LangChain, LangGraph, Hugging Face, Transformers, Prompt Engineering, Embeddings, LoRA Fine-tuning

MLOps and Data: MLflow, FastAPI, Docker, GitHub Actions CI/CD, Git, FAISS, ChromaDB, Pandas, NumPy

Building: DVC, recommendation systems, model serving, evaluation metrics

EDUCATION

Visvesvaraya National Institute of Technology (VNIT), Nagpur

2024 – May 2026

M.Tech, Applied AI and Machine Learning

CGPA: 6.53/10

Dr. D.Y. Patil Institute of Technology, Pune

2019 – 2023

B.E., Computer Science

CGPA: 7.89/10

CERTIFICATIONS

- Coursera: Deep Learning Specialization (in progress)
- DeepLearning.AI: Generative AI with LLMs (in progress)
- Docker Mastery, Udemy (in progress)
- Google Cloud: Cloud Fundamentals

EXPERIENCE

Software Engineer Intern, Salesforce (Remote)

Feb 2022 – May 2022

- Built Python bulk data-processing pipelines, cutting manual data handling by approximately 40%.
- Integrated Salesforce APIs into optimized backend workflows, improving system efficiency by approximately 25%.
- Wrote SQL queries with joins and aggregations to support CRM data analysis for business teams.

PROJECTS

End-to-End MLOps Pipeline for Loan Default Prediction

2026

Python, Scikit-learn, MLflow, DVC, FastAPI, Docker, GitHub Actions

- Built a training pipeline that tracks experiments with MLflow and versions data with DVC.
- Served the best model through a FastAPI endpoint packaged in Docker.
- Automated tests and deployment with GitHub Actions CI/CD.

Product Recommendation System with Embeddings

2026

Python, PyTorch, Sentence Transformers, FAISS, FastAPI

- Built a recommender that combines collaborative filtering with embedding-based similarity search in FAISS.
- Evaluated it with precision@k and recall@k against a popularity baseline, and exposed it through FastAPI.

RAG-Based Clinical Question Answering System

2025

Python, LangChain, ChromaDB, FastAPI, Docker, GitHub Actions

- Built a RAG pipeline for clinical document retrieval with LangChain and ChromaDB, reaching 92% answer relevance on a 100-question benchmark.
- Added hybrid search (BM25 + dense embeddings) with Reciprocal Rank Fusion, improving retrieval accuracy by 18%; deployed with FastAPI, Docker and GitHub Actions CI/CD at under 500 ms latency.

LLM-Powered Security Log Analyzer

2025

Python, LangGraph, OpenAI API

- Built a LangGraph agent workflow with the OpenAI API to analyze 50K+ security logs and generate natural-language threat summaries.
- Added a self-reflection evaluation loop that kept hallucination under 3% on 200 curated test cases.

Military Vehicle Detection and Face Authentication Security System

Academic Project

Python, YOLOv8, OpenCV, Haar Cascade

- Built a real-time surveillance system with Haar Cascade face authentication at approximately 95% detection accuracy.
- Detected and tracked military vehicles with YOLOv8 and OpenCV, sending automated email alerts for security events.

Brain Tumor Detection from MRI Scans

2024

Python, Deep Learning

- Trained a deep learning model to classify brain tumors on 3,000+ MRI scans.

Mobile Botnet Detection System

2023

Python, Scikit-learn

- Built a machine learning model to detect malicious bot communications with C&C servers by analyzing Android network traffic patterns.